

# H2-TRIAD

Interface-Engineered Fe-Ni/N-C/CaF<sub>2</sub> Catalysis for Reversible MgH<sub>2</sub> Hydrogen Storage

Integrated Catalyst Concept and Literature-Grounded Digital Twin

Live Demo: [hydrax-h2-triad.up.railway.app](https://hydrax-h2-triad.up.railway.app) Repository:  
[github.com/YahyaAlsharif/h2-triad](https://github.com/YahyaAlsharif/h2-triad)

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## Executive Summary

H2-TRIAD combines a falsifiable catalyst-development concept with an implemented, evidence-conscious Digital Twin. The material hypothesis is a deliberately pre-assembled Fe-Ni/N-C/CaF<sub>2</sub> interface: Fe-Ni nanoparticles and CaF<sub>2</sub> nanodomains are spatially anchored on an N-doped porous-carbon scaffold before the catalyst is integrated with MgH<sub>2</sub>. The novelty claim is therefore the engineered interface and its validation strategy, not the mere presence of familiar components.

The decisive laboratory comparison is S4, the pre-assembled architecture, versus S3, an equal-composition physical mixture prepared and tested under identical conditions. Fe-Ni has strong precedent as a kinetic promoter, N-C can support dispersion and transport, and fluoride-containing systems motivate interfacial study. However, CaF<sub>2</sub> is treated as a testable modifier, not as a proven route to lower the thermodynamic enthalpy of MgH<sub>2</sub>. Any thermodynamic claim requires pressure-composition-temperature data and van't Hoff analysis.

The software prototype supports this program by preserving literature provenance, returning exact measurements before model estimates, rejecting unsupported inputs, and exposing uncertainty. Its current evidence base contains 119 accessible literature observations; 109 observations from 16 papers and 36 samples enter a bounded CatBoost capacity model. Paper-held-out evaluation gives a mean absolute error of 1.1693 wt.% H<sub>2</sub> and  $R^2 = 0.3314$ . These are actual software-evaluation results, not measurements of S4. The proposed numerical catalyst thresholds are design targets to be accepted, revised, or rejected experimentally.

**Submission claim: H2-TRIAD is a working digital research aid and a proposed catalyst architecture. No laboratory performance is claimed for the Fe-Ni/N-C/CaF<sub>2</sub> interface.**

## 1 Challenge and Project Objective

MgH<sub>2</sub> is attractive because magnesium is abundant and the hydride offers a theoretical material-level hydrogen capacity of about 7.6 wt.%. Its practical limitations include strong Mg-H thermodynamics, slow molecular-hydrogen activation at the surface, slow solid-state transport, and poor heat transfer. Pure or lightly processed MgH<sub>2</sub> therefore commonly requires elevated temperatures and long reaction times [1, 2].

The project objective is to develop and validate a noble-metal-free, interface-engineered Fe-Ni/N-C/CaF<sub>2</sub> catalyst that accelerates reversible absorption and desorption, preserves useful composite capacity, and resists agglomeration over cycling. Kinetic improvement and any genuine thermodynamic change will be evaluated separately. Lower onset temperature by itself will not be interpreted as proof of a lower reaction enthalpy.

## 2 Concept and Candidate Novelty

The final candidate material is MgH<sub>2</sub> combined with a *pre-assembled Fe-Ni/N-C/CaF<sub>2</sub> interface architecture*. The first comparison materials are controls, not competing project ideas. The

central hypothesis is that co-locating the three functions creates persistent reaction boundaries and shorter transport paths that cannot be reproduced reliably by random contact after milling.

|                                                                                           |                                                         |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Fe-Ni</b>                                                                              | H <sub>2</sub> activation and catalytic phase evolution |
| <b>N-C</b>                                                                                | particle dispersion and short transport paths           |
| <b>CaF<sub>2</sub></b>                                                                    | testable interfacial modification and nanoscale spacing |
| ↓ spatially co-located before integration ↓                                               |                                                         |
| <b>PRE-ASSEMBLED Fe-Ni/N-C/CaF<sub>2</sub> INTERFACE</b> → MgH <sub>2</sub> storage phase |                                                         |
| Conceptual architecture, not to scale                                                     |                                                         |

## 2.1 Functional division and evidence status

**Table 1:** Proposed functions are separated from the evidence needed to establish them.

| Element                      | Primary proposed role                                                                                                           | Evidence boundary and decisive check                                                                                                                                                                                          |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fe-Ni nanoparticles          | Promote H <sub>2</sub> dissociation/recombination and form catalytically active Fe- and Mg-Ni-containing phases during cycling. | Supported by Fe-Ni/carbon studies in Mg/MgH <sub>2</sub> systems [3, 4]. Verify phase evolution, apparent activation energy, and identical-condition isothermal rates.                                                        |
| N-doped carbon (N-C)         | Disperse nanoparticles, limit agglomeration, and provide short transport paths.                                                 | N-doped porous-carbon confinement has demonstrated kinetic and cycling benefits in a different MgH <sub>2</sub> architecture [5]. Verify distribution, porosity, and post-cycle structure.                                    |
| CaF <sub>2</sub> nanodomains | Hypothesized ionic/defect-interface modifier and nanoscale spacer at the catalyst-MgH <sub>2</sub> boundary.                    | Fluoride-containing additives and other CaF <sub>2</sub> chemistries provide adjacent precedent, not proof for this architecture [6, 7, 8]. Verify chemical state, interface persistence, kinetic barriers, and PCT behavior. |
| Pre-assembled architecture   | Place the three functions at the same reaction boundary before mixing with MgH <sub>2</sub> .                                   | Project’s candidate novelty. Isolate its effect by comparing S4 with the equal-composition physical mixture S3.                                                                                                               |

The project does not claim that MgH<sub>2</sub>, Fe-Ni catalysis, carbon supports, N-C, or fluoride additives are individually new. A preliminary scan not finding the identical architecture would support further investigation, but it would not constitute a legal patentability opinion.

## 3 Evidence Classes and Claim Discipline

**Established literature evidence** retains the cited system’s composition and conditions; it supports plausibility but does not prove the proposed architecture.

**Proposed mechanisms** are falsifiable explanations, including co-location effects, defect or ionic modification near CaF<sub>2</sub>, and stabilization of Fe-Ni sites.

**Design targets** are go/no-go thresholds, not predictions or achieved values.

**Actual project results** currently comprise Digital Twin implementation and evaluation only; H2-TRIAD has no laboratory measurements for S0–S4 and claims no optimized catalyst.

## 4 Implemented Digital Twin

The Digital Twin is a literature-grounded predictive workspace for condition-aware comparison of  $\text{MgH}_2$  experiments. Its scientific decision rule is:

**Exact literature measurement  $\rightarrow$  supported AI prediction  $\rightarrow$  unavailable**

For a verified sample at matching recorded conditions, the system returns the exact literature measurement with its qualifier and provenance. If no exact measurement exists, the request may be evaluated by the capacity model only after numeric and chemistry support checks. A supported prediction includes an empirical interval, model identity, and warnings. Outside the documented domain, the response is *unavailable* and contains no fabricated capacity. Missing conditions are not wildcards, distinct measurements are not averaged into synthetic observations, and unavailable does not mean zero.

The screenshot displays the 'Hydrogen materials workspace' interface. On the left, the 'Configure an experiment' section includes fields for 'Measurement mode' (set to 'absorption'), 'Duration (s)' (1000), 'Temperature (°C)' (75), 'Catalyst loading (wt.%)' (10), 'Catalyst family' (carbon supported metal), 'Support material' (mesoporous carbon), 'Catalyst components' (C/Ni), 'Catalyst elements' (C/Ni), 'Hydrogen pressure (bar)' (20), and 'Pressure relation' (set to '='). A 'Reset' button is located at the top right of this section. On the right, the 'Capacity result' section shows an 'AI prediction' of 3.61 wt.% H<sub>2</sub>. Below this, it indicates an empirical interval of 1.16-6.70 wt.% H<sub>2</sub>. A note at the bottom states: 'H<sub>2</sub> load capacity > 1 - within the documented support rules. Intervals reflect literature field-of-view, with 86.2% observed coverage. They are not a confidence score or guarantee.'

**Figure 1:** Implemented H2-TRIAD workflow for a supported absorption estimate. Submitted conditions, model output, empirical interval, and support status remain visible together.

## 5 Scientific Data and Predictive Model

The read-only scientific index joins canonical source, sample, and measurement records. Its 119 accessible ordinary absorption/desorption observations retain source identity, sample and preparation context, conditions, capacity basis, qualifiers, and source locators. Approximate and threshold values remain qualified, and reported room temperature is not converted into an invented number. Historical synthetic/demo records remain in a separate SQLite dataset and do not enter literature lookup or model training.

The capacity-training view contains 109 usable scalar observations from 16 independent papers and 36 samples. The compact CatBoost regressor uses measurement mode, temperature, duration, catalyst loading, catalyst presence and component count, and support context. Chemistry and pressure also participate in eligibility checks. Evaluation uses leave-one-paper-out grouping so correlated rows from one publication cannot enter both training and testing.

**Table 2:** Current data and software-evaluation results.

| Item                               | Current value              |
|------------------------------------|----------------------------|
| Accessible literature observations | 119                        |
| Model-training observations        | 109                        |
| Independent training papers        | 16                         |
| Training samples                   | 36                         |
| Paper-held-out MAE                 | 1.1693 wt.% H <sub>2</sub> |
| Paper-held-out RMSE                | 1.5493 wt.% H <sub>2</sub> |
| Paper-held-out $R^2$               | 0.3314                     |
| Empirical 90% interval coverage    | 86.2%                      |
| Mean interval width                | 5.0876 wt.% H <sub>2</sub> |

The signed-residual empirical interval is derived from grouped cross-validation. Its 86.2% observed coverage is not a formal guarantee, confidence score, or laboratory uncertainty. The broad mean interval is itself evidence that the model is data-limited. Support rules reject out-of-range inputs and unseen chemistry, but being inside independent feature ranges does not establish dense joint experimental coverage.

## 6 Material Route and Controlled Validation

Prepare an N-doped porous-carbon scaffold, anchor nanoscale Fe-Ni at its active sites, introduce spatially dispersed CaF<sub>2</sub> domains, and integrate the pre-assembled catalyst with MgH<sub>2</sub> by controlled milling under inert atmosphere. Fe:Ni ratio, total catalyst loading, CaF<sub>2</sub> fraction, scaffold properties, and milling energy are optimization variables rather than fixed claims.

### 6.1 S0–S4 control logic

**Table 3:** Control matrix for isolating component and architecture effects.

| Code | Material                                                              | Question answered                                                    |
|------|-----------------------------------------------------------------------|----------------------------------------------------------------------|
| S0   | Ball-milled MgH <sub>2</sub>                                          | What is the baseline under the same preparation and test conditions? |
| S1   | MgH <sub>2</sub> + CaF <sub>2</sub>                                   | Does CaF <sub>2</sub> have an independent effect?                    |
| S2   | MgH <sub>2</sub> + Fe-Ni/N-C                                          | What is achieved without CaF <sub>2</sub> ?                          |
| S3   | MgH <sub>2</sub> + Fe-Ni/N-C + CaF <sub>2</sub> as a physical mixture | Does simple addition of all ingredients help?                        |
| S4   | MgH <sub>2</sub> + pre-assembled Fe-Ni/N-C/CaF <sub>2</sub>           | Does the engineered interface outperform the same composition?       |

**Decisive validation: S4 must outperform S3 under identical composition, preparation history, sample mass, temperature, pressure, and analysis method. S4 versus S0 alone cannot establish interface synergy.**

## 6.2 Measurements and decision logic

**Table 4:** Measurements needed to distinguish kinetics, thermodynamics, and durability.

| Measurement                       | Decision supported                                                                                                                   |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| XRD, XPS, and TEM/EDS             | Identify phases and chemical states, quantify dispersion, and determine whether the designed interface survives milling and cycling. |
| Sieverts isothermal tests         | Measure reversible capacity and time to 90% absorption/desorption at identical temperature, pressure, and sample mass.               |
| DSC/TGA at multiple heating rates | Compare onset/peak temperature and estimate apparent activation energy with one consistent method.                                   |
| PCT at three or more temperatures | Determine equilibrium pressure and van’t Hoff enthalpy/entropy, separating equilibrium thermodynamics from faster kinetics.          |
| Repeated cycling and replicates   | Quantify capacity/rate retention, uncertainty, phase growth, and catalyst agglomeration.                                             |

## 6.3 Proposed go/no-go targets

These thresholds guide experimental design; none is a reported or model-predicted S4 result.

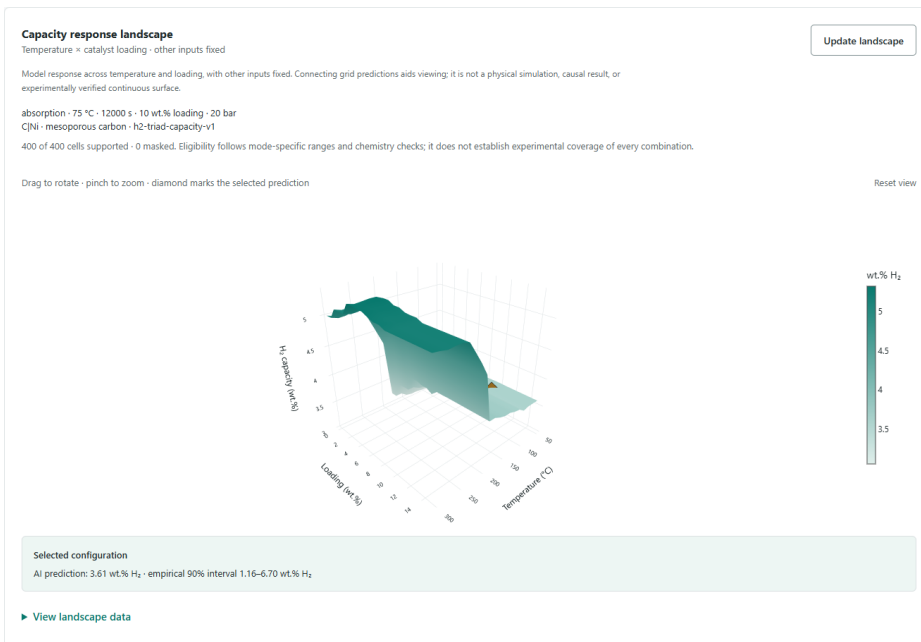
- S4 reduces time to 90% conversion by at least 50% versus both S2 and S3 at identical conditions.
- Desorption onset is at or below 250 °C, or at least 75 °C below S0 under the same heating protocol.
- Reversible composite capacity is at least 5.5 wt.% H<sub>2</sub> with total catalyst loading at or below 10 wt.%.
- At least 90% of initial reversible capacity remains after 50 cycles, with replicate testing and uncertainty reported.
- A statistically significant architecture or component-interaction effect is observed; otherwise, the synergy claim is rejected.

## 7 Computational and AI Support

Computational work is an experiment-guidance layer, not replacement evidence. DFT can compare hydrogen adsorption/dissociation, vacancy formation, charge transfer, and diffusion barriers in simplified MgH<sub>2</sub>/Fe-Ni, MgH<sub>2</sub>/N-C, MgH<sub>2</sub>/CaF<sub>2</sub>, and combined-interface models. Data-driven barrier models demonstrate how lower-cost descriptors can prioritize calculations, but their predictions still require experimental confrontation [9].

After measured S0–S4 data exist, a small-data Bayesian optimizer can recommend the next Fe:Ni ratio, catalyst loading, CaF<sub>2</sub> fraction, or milling condition. Each plotted curve must be labeled as a measured result, literature benchmark, or model prediction. The current CatBoost model explores capacity trends in the curated literature; it does not predict that the proposed S4 architecture will meet the targets above.

For supported configurations, the deployed application can evaluate a temperature-by-loading grid while holding other inputs fixed. Every cell is screened before inference; unsupported cells remain null and are excluded from the surface.



**Figure 2:** Implemented response landscape. It visualizes learned CatBoost behavior over supported cells; it is not a molecular simulation, causal result, or experimentally verified continuous surface.

## 8 System Architecture and Current Validation

H2-TRIAD uses a React/Vite frontend and FastAPI backend. The backend connects the interface to canonical literature tables and a cached CatBoost predictor for scientific results, while a separate SQLite store supports the explicitly labeled historical synthetic/demo explorer. A repository-root Dockerfile builds the frontend and serves the bundled application and `/api` routes through one Uvicorn process on Railway.

Release validation currently records 22 model tests, 72 backend tests, and 26 browser tests, including the integrated container workflow. Checks cover exact-measurement precedence, qualified values, unavailable responses without numeric fields, landscape masking, scientific dataset integrity, and stable production artifacts. This establishes a working software prototype, not a validated physical catalyst.

## 9 Hackathon Demonstrator, Value, and Risk

The demonstration links the implemented software to the next experimental decision. A judge can configure a condition-aware  $\text{MgH}_2$  comparison, inspect an exact literature result or bounded model estimate, review its provenance and uncertainty, and then examine the S0-S4 validation plan. A physical display can use a sealed cartridge cutaway and non-reactive material mock-up; no live hydrogen is required.

The likely first application context is stationary storage, renewable buffering, or a heat-integrated system. The concept avoids noble metals and uses broadly available Fe, Ni, carbon, and  $\text{CaF}_2$ , but nanostructure processing, inert handling, oxidation control, and heat management still require technical and economic assessment. High heat demand may limit mobile use, so no vehicle-readiness claim is made.

Real powder synthesis and pressure testing must occur only in a qualified laboratory with inert handling, approved pressure equipment, ventilation, hydrogen detection, and documented risk review.

## 10 Limitations

- The exact pre-assembled Fe-Ni/N-C/CaF<sub>2</sub> architecture has not been synthesized or tested by this project.
- Evidence for CaF<sub>2</sub> is indirect and composition-dependent; inactivity, adverse phases, or capacity dilution remain plausible outcomes.
- The literature dataset is small at the independent-paper level and concentrated in particular catalyst families.
- Paper-held-out performance is moderate, and broad empirical intervals expose substantial uncertainty.
- Feature-wise support checks do not prove joint experimental coverage, causal optimization, or mechanistic validity.
- The exact combination’s patentability is unconfirmed pending a formal prior-art review.

## 11 Conclusion

H2-TRIAD advances a coherent hackathon proposition: a deliberately pre-assembled Fe-Ni/N-C/CaF<sub>2</sub> interface to be tested through a decisive S4-versus-S3 control, paired with a working Digital Twin that keeps literature observations, bounded predictions, uncertainty, and unsupported regions visibly distinct. The proposal is scientifically valuable precisely because it is falsifiable. If S4 does not outperform the equal-composition mixture, the interface-synergy claim is rejected; if equilibrium measurements do not change, the system remains correctly described as a kinetic catalyst. Computational and AI tools prioritize experiments, while laboratory measurements retain the final evidentiary authority.

## References

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